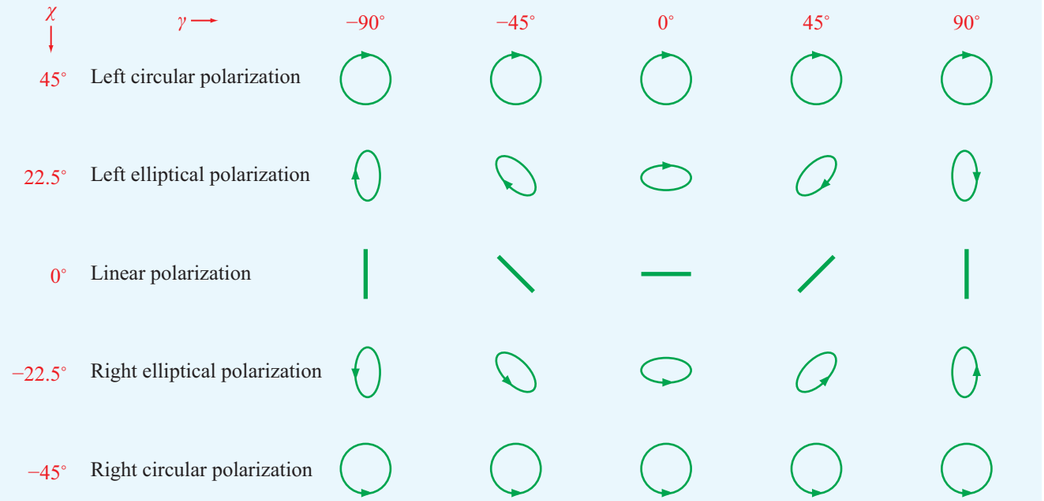


Figure 7-12 Polarization states for various combinations of the polarization angles (γ, χ) for a wave traveling out of the page.

where we have replaced the negative sign of the second term with e^{j180° in order to have positive amplitudes for both terms, thereby allowing us to use the definitions given earlier. According to the expression for $\widetilde{\mathbf{E}}(z)$, the phase angles of the x and y components are $\delta_x = 30^\circ$ and $\delta_y = 135^\circ$, giving a phase difference of $\delta = \delta_y - \delta_x = 135^\circ - 30^\circ = 105^\circ$. The auxiliary angle ψ_0 is obtained from

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^\circ.$$

From Eq. (7.59a),

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta = \tan 106.2^\circ \cos 105^\circ = 0.89,$$

which gives two solutions for γ, namely $\gamma = 20.8^\circ$ and $\gamma = -69.2^\circ$. Since $\cos\delta < 0$, the correct value of γ is -69.2°. From Eq. (7.59b),

$$\sin 2\chi = (\sin 2\psi_0)\sin\delta$$
$$= \sin 106.2^\circ \sin 105^\circ = 0.93 \quad \text{or} \quad \chi = 34.0^\circ.$$

The magnitude of χ indicates that the wave is elliptically polarized, and its positive polarity specifies its rotation as left handed.

Concept Question 7-4: An elliptically polarized wave is characterized by amplitudes a_x and a_y and by the phase difference δ. If a_x and a_y are both nonzero, what should δ be in order for the polarization state to reduce to linear polarization?

Concept Question 7-5: Which of the following two descriptions defines an RHC-polarized wave: A wave incident upon an observer is RHC-polarized if its electric field appears to the observer to rotate in a counterclockwise direction (a) as a function of time in a fixed plane perpendicular to the direction of wave travel or (b) as a function of travel distance at a fixed time t?

Exercise 7-5: The electric field of a plane wave is given by

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,3\cos(\omega t - kz) + \hat{\mathbf{y}}\,4\cos(\omega t - kz) \quad \text{(V/m)}.$$

Determine (a) the polarization state, (b) the modulus of $\mathbf{E}$, and (c) the auxiliary angle.

Answer: (a) Linear, (b) $|\mathbf{E}| = 5\cos(\omega t - kz)$ (V/m), (c) $\psi_0 = 53.1^\circ$. (See EM.)

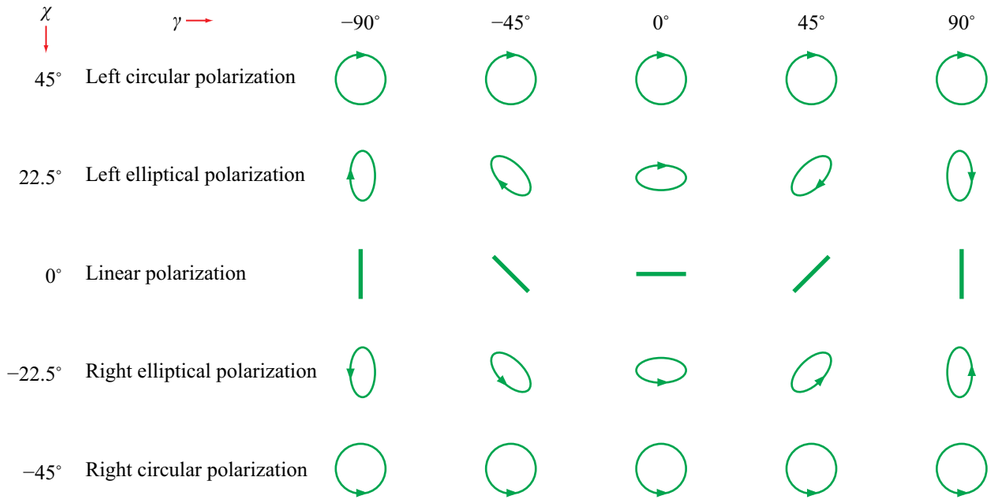


图 7-12 传播到页面之外的波的偏振角 (γ、χ) 的各种组合的偏振状态。

其中我们用 e^{j180° 替换了第二项的负号，以便两项都有正振幅，从而允许我们使用前面给出的定义。根据 $\widetilde{E}(z)$ 的表达式，x和y分量的相位角为 $\delta_x = 30^\circ$ 和 $\delta_y = 135^\circ$，相位差为 $\delta = \delta_y - \delta_x = 135^\circ - 30^\circ = 105^\circ$。辅助角 ψ_0 由下式获得

$$\psi_0 = \tan^{-1}\left(\frac{a_y}{a_x}\right) = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^\circ.$$

从方程。(7.59a),

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta = \tan 106.2^\circ \cos 105^\circ = 0.89,$$

它给出了 γ 的两个解，即 $\gamma = 20.8^\circ$ 和 $\gamma = -69.2^\circ$。由于 $\cos\delta < 0$，γ 的正确值为 -69.2°。从方程。(7.59b),

$$\sin 2\chi = (\sin 2\psi_0)\sin\delta$$
$$= \sin 106.2^\circ \sin 105^\circ = 0.93 \quad \text{or} \quad \chi = 34.0^\circ.$$

χ 的大小表明波是椭圆偏振的，其正极性指定其旋转为左手。

概念问题 7-4：椭圆偏振波的特征是振幅 a_x 和 a_y 以及相位差 δ。如果 a_x 和 a_y 都不为零，那么 δ 应该是多少才能使偏振态降低为线偏振？

概念问题 7-5：以下哪两个描述定义了 RHC 偏振波：如果观察者认为入射到观察者的波的电场沿逆时针方向旋转，则该波是 RHC 偏振的（a）作为垂直于波传播方向的固定平面中时间的函数，或（b）作为固定时间 t 处传播距离的函数？

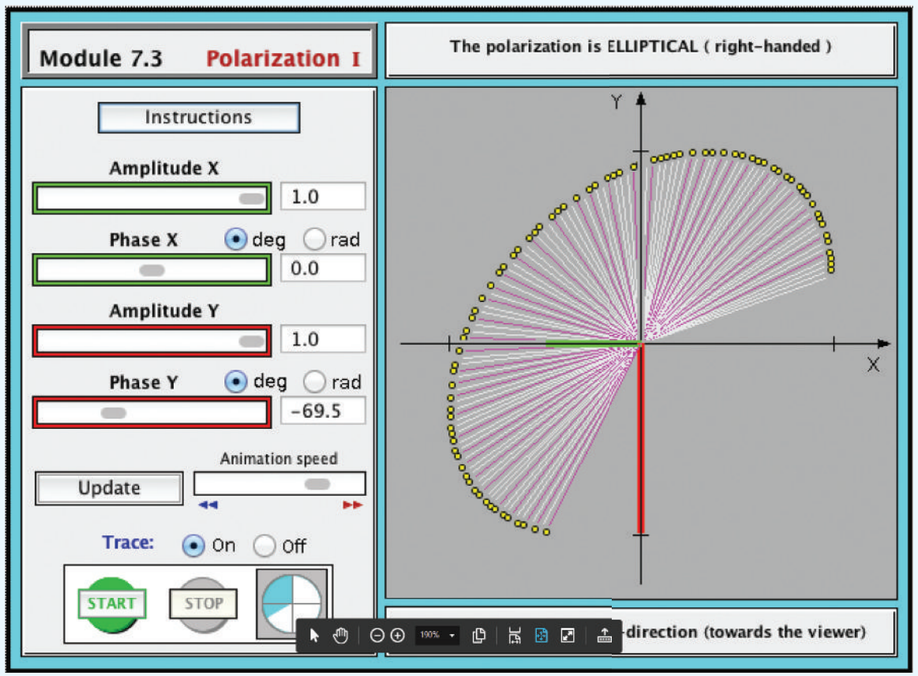
练习7-5：平面波的电场由下式给出

$$\mathbf{E}(z,t) = \hat{\mathbf{x}}\,3\cos(\omega t - kz) + \hat{\mathbf{y}}\,4\cos(\omega t - kz) \quad \text{(V/m)}.$$

确定 (a) 偏振态、(b) E 模量和 (c) 辅助角。

答案：(a) 线性，(b) $|E| = 5\cos(\omega t - kz)$ (V/m)，(c) $\psi_0 = 53.1^\circ$。（参见EM。）

Module 7.3 Polarization I Upon specifying the amplitudes and phases of the x and y components of $\mathbf{E}$, the user can observe the trace of $\mathbf{E}$ in the x–y plane.



Exercise 7-6: If the electric field phasor of a TEM wave is given by $\widetilde{\mathbf{E}} = (\hat{\mathbf{y}} - \hat{\mathbf{z}}j)e^{-jkx}$, determine the polarization state.

Answer: RHC polarization. (See EM.)

7-4 Plane-Wave Propagation in Lossy Media

To examine wave propagation in a lossy (conducting) medium, we return to the wave equation given by Eq. (7.15),

$$\nabla^2\widetilde{\mathbf{E}} - \gamma^2\widetilde{\mathbf{E}} = 0 \tag{7.61}$$

with

$$\gamma^2 = -\omega^2\mu\varepsilon_c = -\omega^2\mu(\varepsilon' - j\varepsilon''), \tag{7.62}$$

where $\varepsilon' = \varepsilon$ and $\varepsilon'' = \sigma/\omega$. Since γ is complex, we express it as

$$\gamma = \alpha + j\beta, \tag{7.63}$$

where α is the medium's ***attenuation constant*** and β its ***phase constant***. By replacing γ with $(\alpha + j\beta)$ in Eq. (7.62), we obtain

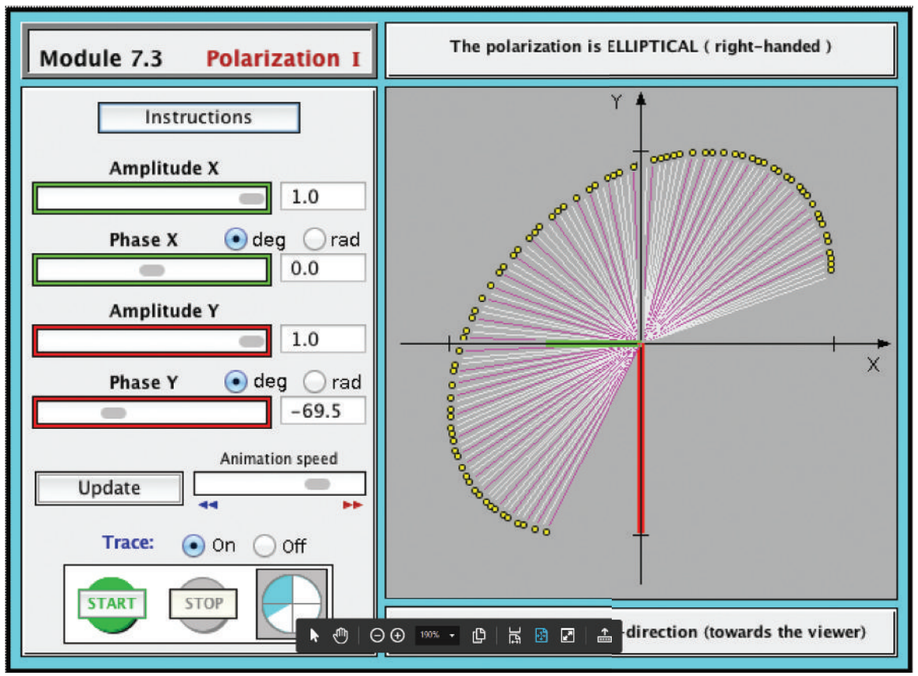
$$(\alpha + j\beta)^2 = (\alpha^2 - \beta^2) + j2\alpha\beta = -\omega^2\mu\varepsilon' + j\omega^2\mu\varepsilon''. \tag{7.64}$$

The rules of complex algebra require the real and imaginary parts on one side of an equation to equal, respectively, the real and imaginary parts on the other side. Hence,

$$\alpha^2 - \beta^2 = -\omega^2\mu\varepsilon', \tag{7.65a}$$

$$2\alpha\beta = \omega^2\mu\varepsilon''. \tag{7.65b}$$

模块 7.3 极化 I 在指定 E 的 x 和 y 分量的幅度和相位后，用户可以观察 E 在 x-y 平面上的迹线。



练习 7-6：如果 TEM 波的电场相量由 $\widetilde{\mathrm{E}} = (\hat{\mathrm{y}} - \hat{\mathrm{z}}\,\mathrm{j})\mathrm{e}^{-jkx}$ 给出，请确定偏振态。

答案：RHC 偏振。 （参见EM。）

7-4 有损介质中的平面波传播

为了检查波在有损（传导）介质中的传播，我们回到方程式给出的波动方程。 （7.15），

$$\nabla^2\widetilde{\mathbf{E}} - \gamma^2\widetilde{\mathbf{E}} = 0 \tag{7.61}$$

和

$$\gamma^2 = -\omega^2\mu\varepsilon_c = -\omega^2\mu(\varepsilon' - j\varepsilon''), \tag{7.62}$$

其中 $\varepsilon' = \varepsilon$ 和 $\varepsilon'' = \sigma/\omega$。由于 γ 很复杂，我们将其表示为

$$\gamma = \alpha + j\beta, \tag{7.63}$$

其中 α 是介质的衰减常数，β 是其相位常数。将等式中的 γ 替换为 $(\alpha + \mathrm{j}\beta)$。 （7.62），我们得到

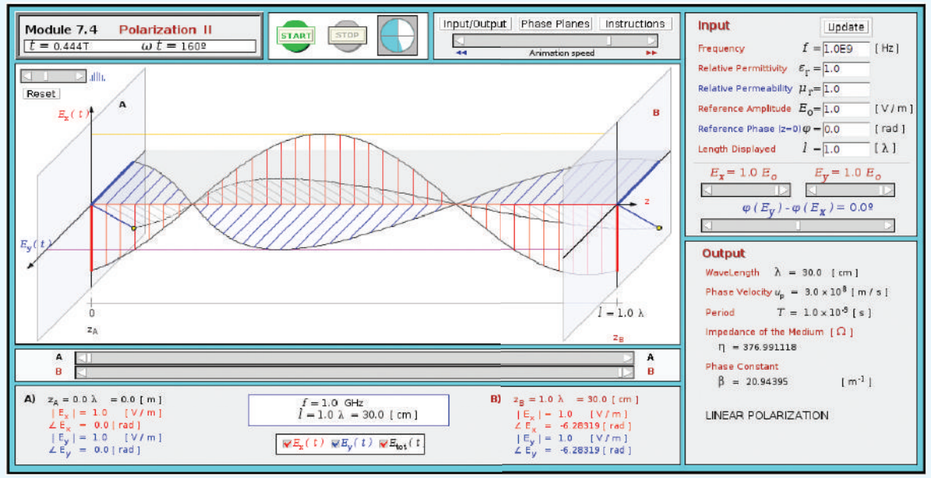
$$(\alpha + j\beta)^2 = (\alpha^2 - \beta^2) + j2\alpha\beta = -\omega^2\mu\varepsilon' + j\omega^2\mu\varepsilon''. \tag{7.64}$$

复代数规则要求方程一侧的实部和虚部分别等于另一侧的实部和虚部。因此，

$$\alpha^2 - \beta^2 = -\omega^2\mu\varepsilon', \tag{7.65a}$$

$$2\alpha\beta = \omega^2\mu\varepsilon''. \tag{7.65b}$$

Module 7.4 Polarization II Upon specifying the amplitudes and phases of the x and y components of $\mathbf{E}$, the user can observe the 3-D profile of the $\mathbf{E}$ vector over a specified length span.



Solving these two equations for α and β gives

$$\alpha = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right] \right\}^{1/2} \quad \text{(Np/m)}, \tag{7.66a}$$

$$\beta = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right] \right\}^{1/2} \quad \text{(rad/m)}. \tag{7.66b}$$

For a uniform plane wave with electric field $\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\,\widetilde{E}_x(z)$ traveling along the z direction, the wave equation given by Eq. (7.61) reduces to

$$\frac{d^2\widetilde{E}_x(z)}{dz^2} - \gamma^2 \widetilde{E}_x(z) = 0. \tag{7.67}$$

The general solution of the wave equation given by Eq. (7.67) comprises two waves: one traveling in the $+z$ direction and another traveling in the $-z$ direction. Assuming only the former is present, the solution of the wave equation leads to

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x(z) = \hat{\mathbf{x}}E_{x0}e^{-\gamma z} = \hat{\mathbf{x}}E_{x0}e^{-\alpha z}e^{-j\beta z}. \tag{7.68}$$

The associated magnetic field $\widetilde{\mathbf{H}}$ can be determined by applying Eq. (7.2b): $\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}$, or using Eq. (7.39a): $\widetilde{\mathbf{H}} = (\hat{\mathbf{k}} \times \widetilde{\mathbf{E}})/\eta_c$, where η_c is the ***intrinsic impedance of the lossy medium***. Both approaches give

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\,\widetilde{H}_y(z) = \hat{\mathbf{y}}\,\frac{\widetilde{E}_x(z)}{\eta_c} = \hat{\mathbf{y}}\,\frac{E_{x0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}, \tag{7.69}$$

where

$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2} \quad (\Omega). \tag{7.70}$$

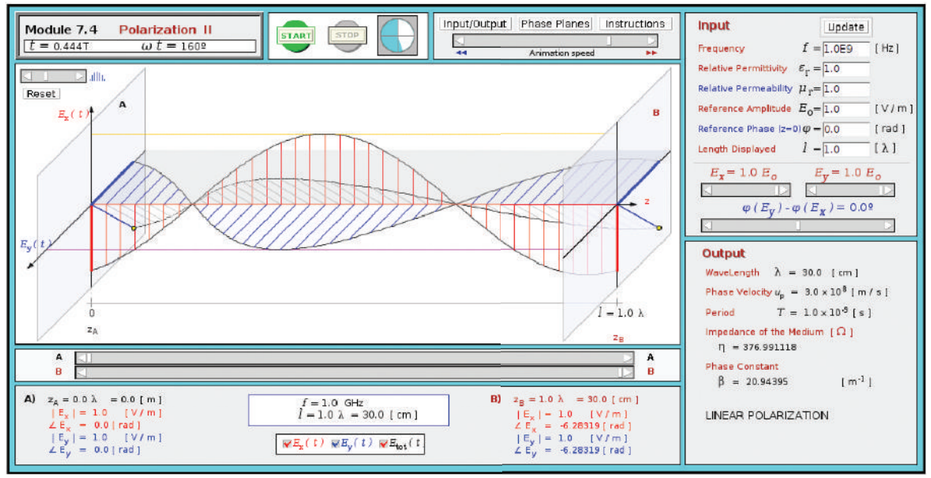
We noted earlier that in a lossless medium, $\mathbf{E}(z,t)$ is in phase with $\mathbf{H}(z,t)$. This property no longer holds true in a lossy medium because η_c is complex. This fact is demonstrated later in Example 7-4.

From Eq. (7.68), the magnitude of $\widetilde{E}_x(z)$ is given by

$$|\widetilde{E}_x(z)| = |E_{x0}e^{-\alpha z}e^{-j\beta z}| = |E_{x0}|e^{-\alpha z}, \tag{7.71}$$

which decreases exponentially with z at a rate dictated by the attenuation constant α. Since $\widetilde{H}_y = \widetilde{E}_x/\eta_c$, the magnitude of $\widetilde{H}_y$ also decreases as $e^{-\alpha z}$. As the field attenuates, part of the energy carried by the electromagnetic wave is converted into

模块 7.4 极化 II 在指定 E 的 x 和 y 分量的幅度和相位后，用户可以在指定的长度范围内观察 E 矢量的 3-D 轮廓。



求解这两个方程的 α 和 β 给出

$$\alpha = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} - 1 \right] \right\}^{1/2} \quad \text{(Np/m)}, \tag{7.66a}$$

$$\beta = \omega \left\{ \frac{\mu\varepsilon'}{2} \left[\sqrt{1 + \left(\frac{\varepsilon''}{\varepsilon'} \right)^2} + 1 \right] \right\}^{1/2} \quad \text{(rad/m)}. \tag{7.66b}$$

对于沿 z 方向传播的电场 $\widetilde{\mathrm{E}} = \hat{\mathrm{x}}\,\widetilde{\mathrm{E}}_x(z)$ 的均匀平面波，波动方程由式（1）给出。(7.61) 减少为

$$\frac{d^2\widetilde{E}_x(z)}{dz^2} - \gamma^2 \widetilde{E}_x(z) = 0. \tag{7.67}$$

方程给出的波动方程的通解。(7.67) 式包含两个波：一个沿 +z 方向传播，另一个沿 −z 方向传播。假设仅存在前者，则波动方程的解导致

$$\widetilde{\mathbf{E}}(z) = \hat{\mathbf{x}}\widetilde{E}_x(z) = \hat{\mathbf{x}}E_{x0}e^{-\gamma z} = \hat{\mathbf{x}}E_{x0}e^{-\alpha z}e^{-j\beta z}. \tag{7.68}$$

相关的磁场 $\widetilde{\mathrm{H}}$ 可以通过应用等式来确定。(7.2b): $\nabla \times \widetilde{\mathrm{E}} = -\mathrm{j}\omega\mu\,\widetilde{\mathrm{H}}$，或使用等式：(7.39a): $\widetilde{\mathrm{H}} = (\hat{\mathrm{k}} \times \widetilde{\mathrm{E}})/\eta_c$，其中 η_c 是有损介质的固有阻抗。两种方法都给出

$$\widetilde{\mathbf{H}}(z) = \hat{\mathbf{y}}\,\widetilde{H}_y(z) = \hat{\mathbf{y}}\,\frac{\widetilde{E}_x(z)}{\eta_c} = \hat{\mathbf{y}}\,\frac{E_{x0}}{\eta_c}e^{-\alpha z}e^{-j\beta z}, \tag{7.69}$$

在哪里

$$\eta_c = \sqrt{\frac{\mu}{\varepsilon_c}} = \sqrt{\frac{\mu}{\varepsilon'}}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{-1/2} \quad (\Omega). \tag{7.70}$$

我们之前注意到，在无损介质中，E(z,t) 与 H(z,t) 同相。此属性在有损介质中不再成立，因为 η_c 很复杂。这个事实在后面的例 7-4 中得到了证明。

从方程。(7.68)，$\widetilde{\mathrm{E}}_x(z)$ 的大小由下式给出

$$|\widetilde{E}_x(z)| = |E_{x0}e^{-\alpha z}e^{-j\beta z}| = |E_{x0}|e^{-\alpha z}, \tag{7.71}$$

它随 z 以衰减常数 α 指定的速率呈指数下降。由于 $\widetilde{\mathrm{H}}_y = \widetilde{\mathrm{E}}_x/\eta_c$，$\widetilde{\mathrm{H}}_y$的幅度也随着$\mathrm{e}^{-\alpha z}$而减小。随着磁场衰减，电磁波携带的部分能量转化为

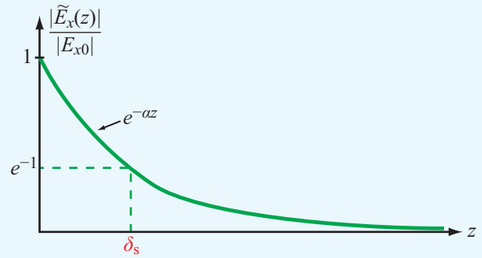


Figure 7-13 Attenuation of the magnitude of $\tilde{E}_x(z)$ with distance z. The skin depth δ_s is the value of z at which $|\tilde{E}_x(z)|/|E_{x0}| = e^{-1}$ or $z = \delta_s = 1/\alpha$.

heat due to conduction in the medium. As the wave travels through a distance $z = \delta_s$ with

$$\delta_s = \frac{1}{\alpha} \quad \text{(m)}, \tag{7.72}$$

the wave magnitude decreases by a factor of $e^{-1} \approx 0.37$ (**Fig. 7-13**). At depth $z = 3\delta_s$, the field magnitude is less than 5% of its initial value, and at $z = 5\delta_s$, it is less than 1%.

► This distance δ_s, called the ***skin depth*** of the medium, characterizes how deep an electromagnetic wave can penetrate into a conducting medium. ◄

In a perfect dielectric, $\sigma = 0$ and $\varepsilon'' = 0$. Use of Eq. (7.66a) yields $\alpha = 0$; therefore, $\delta_s = \infty$. Thus, in free space, a plane wave can propagate indefinitely with no loss in magnitude. On the other extreme, in a perfect conductor, $\sigma = \infty$ and use of Eq. (7.66a) leads to $\alpha = \infty$. Hence $\delta_s = 0$. If the outer conductor of a coaxial cable is designed to be several skin depths thick, it prevents energy inside the cable from leaking outward and shields against penetration of electromagnetic energy from external sources into the cable.

The expressions given by Eqs. (7.66a), (7.66b), and (7.70) for α, β, and η_c are valid for any linear, isotropic, and homogeneous medium. For a perfect dielectric ($\sigma = 0$), these expressions reduce to those for the lossless case (Section 7-2), wherein $\alpha = 0$, $\beta = k = \omega\sqrt{\mu\varepsilon}$, and $\eta_c = \eta$. For a lossy medium, the ratio $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon$, which appears in all these expressions, plays an important role in classifying how lossy the medium is. When $\varepsilon''/\varepsilon' \ll 1$, the medium is considered a ***low-loss dielectric***, and when $\varepsilon''/\varepsilon' \gg 1$, it is considered a ***good conductor***. In practice, the medium may be regarded as a low-loss dielectric if $\varepsilon''/\varepsilon' < 10^{-2}$, as a good conductor if $\varepsilon''/\varepsilon' > 10^2$, and as a ***quasi-conductor*** if $10^{-2} \le \varepsilon''/\varepsilon' \le 10^2$. For low-loss dielectrics and good conductors, the expressions given by Eq. (7.66) can be significantly simplified, as shown next.

7-4.1 Low-Loss Dielectric

From Eq. (7.62), the general expression for γ is

$$\gamma = j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{1/2}. \tag{7.73}$$

For $|x| \ll 1$, the function $(1-x)^{1/2}$ can be approximated by the first two terms of its binomial series; that is, $(1-x)^{1/2} \approx 1 - x/2$. By applying this approximation to Eq. (7.73) for a low-loss dielectric with $x = j\varepsilon''/\varepsilon'$ and $\varepsilon''/\varepsilon' \ll 1$, we obtain

$$\gamma \approx j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\varepsilon''}{2\varepsilon'}\right). \tag{7.74}$$

The real and imaginary parts of Eq. (7.74) are

$$\alpha \approx \frac{\omega\varepsilon''}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \quad \text{(Np/m)}, \tag{7.75a}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \omega\sqrt{\mu\varepsilon} \quad \text{(rad/m)}. \tag{7.75b}$$

(low-loss medium)

We note that the expression for β is the same as that for the wavenumber k of a lossless medium. Applying the binomial approximation $(1-x)^{-1/2} \approx (1+x/2)$ to Eq. (7.70) leads to

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\varepsilon''}{2\varepsilon'}\right) = \sqrt{\frac{\mu}{\varepsilon}}\left(1 + j\frac{\sigma}{2\omega\varepsilon}\right). \tag{7.76a}$$

In practice, because $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon < 10^{-2}$, the second term in Eq. (7.76a) often is ignored. Thus,

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon}}, \tag{7.76b}$$

which is the same as Eq. (7.31) for the lossless case.

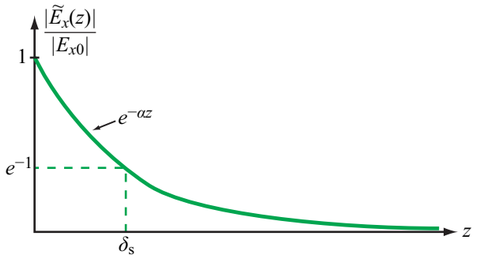


图7-13 $\tilde{E}_x(z)$的幅度随距离z的衰减。趋肤深度 δ_s 是 $|\tilde{E}_x(z)|/|E_{x0}| = e^{-1}$ 或 $z = \delta_s = 1/\alpha$ 处的 z 值。

由于介质传导而产生热量。当波传播距离 $z = \delta_s$ 时

$$\delta_s = \frac{1}{\alpha} \quad \text{(m)}, \tag{7.72}$$

波幅减小了 $e^{-1} \approx 0.37$ 倍（图 7-13）。在深度 $z = 3\delta_s$ 处，场大小小于其初始值的 5%，而在 $z = 5\delta_s$ 处，场大小小于其初始值的 1%。

► 这个距离 δ_s 称为介质的趋肤深度，表征电磁波可以穿透导电介质的深度。 ◄

在理想的电介质中，$\sigma = 0$ 和 $\varepsilon'' = 0$。(7.66a) 产生 $\alpha = 0$；因此，$\delta_s = \infty$。因此，在自由空间中，平面波可以无限传播而不会损失幅度。在另一个极端，在理想导体中，$\sigma = \infty$ 和方程的使用。(7.66a) 导致 $\alpha = \infty$。因此$\delta_s = 0$。如果同轴电缆的外导体被设计为几个趋肤深度厚，它可以防止电缆内部的能量向外泄漏，并屏蔽来自外部源的电磁能量渗透到电缆中。

等式给出的表达式。α、β 和 η_c 的 (7.66a)、(7.66b) 和 (7.70) 对于任何线性、各向同性和均匀介质都有效。对于理想电介质 ($\sigma = 0$)，这些表达式简化为无损情况的表达式（第 7-2 节），其中 $\alpha = 0$、$\beta = k = \omega\sqrt{\mu\varepsilon}$ 和 $\eta_c = \eta$。对于有损介质，出现在所有这些表达式中的比率 $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon$ 在对介质有损程度进行分类时起着重要作用。当$\varepsilon''/\varepsilon' \ll 1$时，考虑介质

一种低损耗电介质，当$\varepsilon''/\varepsilon' \gg 1$时，它被认为是良导体。实际上，如果$\varepsilon''/\varepsilon' < 10^{-2}$，介质可以被视为低损耗电介质，如果$\varepsilon''/\varepsilon' > 10^2$，则介质可以被视为良导体，如果$10^{-2} \le \varepsilon''/\varepsilon' \le 10^2$，则可以被视为准导体。对于低损耗电介质和良导体，方程给出的表达式：(7.66) 可以被大大简化，如下所示。

7-4.1 低损耗电介质

从方程。(7.62)，γ 的一般表达式为

$$\gamma = j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\varepsilon''}{\varepsilon'}\right)^{1/2}. \tag{7.73}$$

对于 $|x| \ll 1$，函数 $(1-x)^{1/2}$ 可以通过其二项式级数的前两项来近似；即 $(1-x)^{1/2} \approx 1 - x/2$。通过将此近似应用于方程。(7.73) 对于 $x = j\varepsilon''/\varepsilon'$ 和 $\varepsilon''/\varepsilon' \ll 1$ 的低损耗电介质，我们得到

$$\gamma \approx j\omega\sqrt{\mu\varepsilon'}\left(1 - j\frac{\varepsilon''}{2\varepsilon'}\right). \tag{7.74}$$

方程的实部和虚部。(7.74) 是

$$\alpha \approx \frac{\omega\varepsilon''}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon}} \quad \text{(Np/m)}, \tag{7.75a}$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \omega\sqrt{\mu\varepsilon} \quad \text{(rad/m)}. \tag{7.75b}$$

(low-loss medium)

我们注意到，β 的表达式与无损介质的波数 k 的表达式相同。将二项式近似 $(1-x)^{-1/2} \approx (1+x/2)$ 应用于等式：(7.70) 导致

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}}\left(1 + j\frac{\varepsilon''}{2\varepsilon'}\right) = \sqrt{\frac{\mu}{\varepsilon}}\left(1 + j\frac{\sigma}{2\omega\varepsilon}\right). \tag{7.76a}$$

实际上，因为 $\varepsilon''/\varepsilon' = \sigma/\omega\varepsilon < 10^{-2}$，方程中的第二项。(7.76a) 经常被忽视。因此，

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon}}, \tag{7.76b}$$

这与等式相同。(7.31) 对于无损情况。

Technology Brief 14: Liquid Crystal Display (LCD)

LCDs are used in digital clocks, cellular phones, desktop and laptop computers, some televisions, and other electronic systems. They offer a decided advantage over former display technologies, such as cathode ray tubes, because they are much lighter and thinner and consume a lot less power to operate. LCD technology relies on special electrical and optical properties of a class of materials known as liquid crystals, which were first discovered in the 1880s by botanist ***Friedrich Reinitzer***.

Physical Principle

► ***Liquid crystals*** are neither a pure solid nor a pure liquid; rather, they are a hybrid of both. ◄

One particular variety of interest is the ***twisted nematic*** liquid crystal whose rod-shaped molecules have a natural tendency to assume a ***twisted spiral structure*** when the material is sandwiched between finely ***grooved glass substrates*** with orthogonal orientations (**Fig. TF14-1**). Note that the molecules in contact with the grooved surfaces align themselves in parallel along the grooves, from a y orientation at the entrance substrate into an x orientation at the exit substrate. The molecular spiral causes the crystal to behave like a ***wave polarizer***: unpolarized light incident upon the entrance substrate follows the orientation of the spiral, emerging through the exit substrate with its polarization (direction of electric field) parallel to the groove's direction, which in **Fig. TF14-1** is along the x direction. Thus, of the x and y components of the incident light, only the y component is allowed to pass through the y-polarized filter, but as a consequence of the spiral action facilitated by the liquid crystal's molecules, the light that emerges from the LCD structure is x-polarized.

LCD Structure

A single-pixel LCD structure is shown in **Fig. TF14-2** for the OFF and ON states, with OFF corresponding to a bright-looking pixel and ON to a dark-looking pixel.

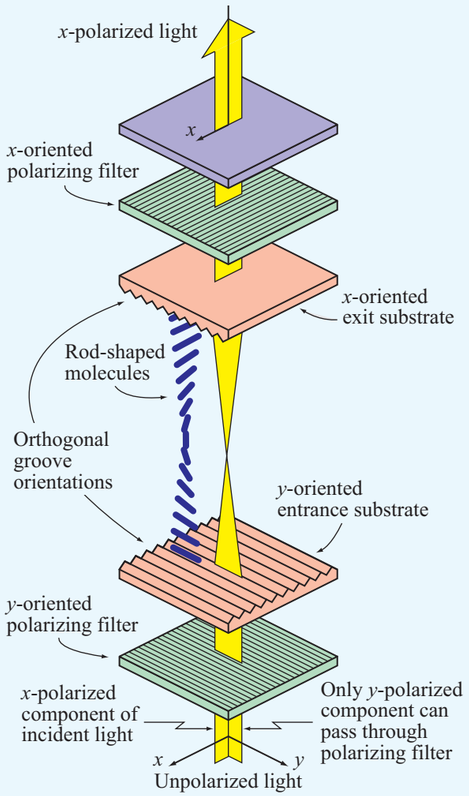


Figure TF14-1 The rod-shaped molecules of a liquid crystal sandwiched between grooved substrates with orthogonal orientations causes the electric field of the light passing through it to rotate by 90°.

► The sandwiched liquid-crystal layer (typically on the order of ***5 microns in thickness*** or 1/20 of the width of a human hair) is straddled by a pair of optical filters with orthogonal polarizations. ◄

When no voltage is applied across the crystal layer (**Fig. TF14-2**(a)), incoming unpolarized light gets polarized as it passes through the entrance polarizer, rotates by 90° as it follows the molecular spiral, and finally

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技术简介 14：液晶显示器 (LCD)

LCD 用于数字时钟、移动电话、台式电脑和笔记本电脑、某些电视以及其他电子系统。与以前的显示技术（例如阴极射线管）相比，它们具有明显的优势，因为它们更轻、更薄，并且运行时消耗的功率也少得多。 LCD 技术依赖于液晶材料的特殊电学和光学特性，这种材料由植物学家 Friedrich Reinitzer 在 1880 年代首次发现。

物理原理

► 液晶既不是纯固体也不是纯液体；相反，它们是两者的混合体。 ◄

一种特别令人感兴趣的品种是扭曲向列液晶，当该材料夹在具有正交取向的精细凹槽玻璃基板之间时，其杆状分子自然倾向于呈现扭曲的螺旋结构（图TF14-1）。请注意，与凹槽表面接触的分子沿着凹槽平行排列，从入口基板处的 y 方向到出口基板处的 x 方向。分子螺旋使晶体表现得像一个波偏振器：入射到入口基板上的非偏振光遵循螺旋的方向，通过出口基板出现，其偏振（电场方向）平行于凹槽的方向，在图 TF14-1 中沿着 x 方向。因此，在入射光的 x 和 y 分量中，只有 y 分量可以通过 y 偏振滤光片，但由于液晶分子促进的螺旋作用，从 LCD 结构中发出的光是 x 偏振的。

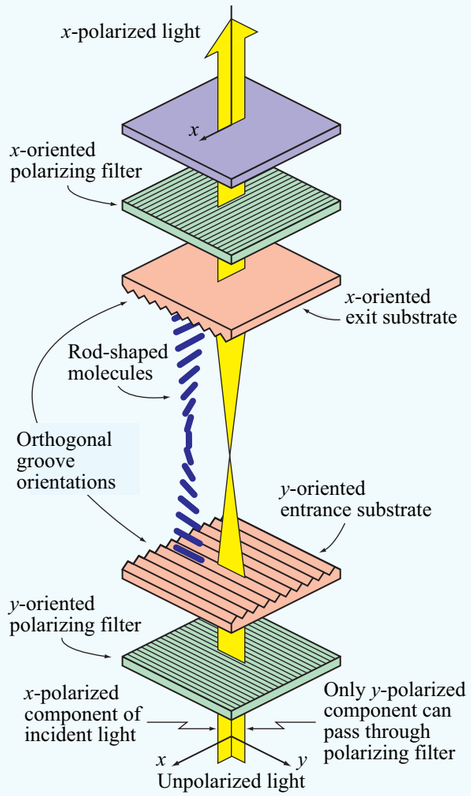


图 TF14-1 液晶的棒状分子夹在具有正交方向的凹槽基板之间，导致穿过它的光的电场旋转 90°。

► 夹在中间的液晶层（厚度通常约为 5 微米或人类头发宽度的 1/20）由一对正交偏振的光学滤光片横跨。 ◄

液晶显示屏结构

图 TF14-2 显示了单像素 LCD 的 OFF 和 ON 状态结构，其中 OFF 对应于亮像素，ON 对应于暗像素。

当晶体层上没有施加电压时（图 TF14-2(a)），入射的非偏振光在穿过入口偏振器时会发生偏振，并沿着分子螺旋旋转 90°，最后